

For data extraction on 7/19/2026 (09:00-)

- **Ingestion run one (first ingestion)**

Table	Records Inserted	Duplicates	Errors
App	5	0	0
Country	5	0	0
Raw_Reviews	10784	0	0
Cleaned_Reviews	8853	0	0

- **Ingestion run two (first ingestion)**

Table	Records Inserted	Duplicates	Errors
App	0	5	0
Country	0	5	0
Raw_Reviews	0	10784	0
Cleaned_Reviews	0	8792	0

-The second ingestion run correctly identified all existing records and skipped duplicate insertions

- **Quality Flags (records needed to be removed)**

Sum(condition=True)

Duplicate Records	Missing Review Content	Non-English Review Content	Empty Review After Cleaning
0	0	1931	0

Missing Review Content

-Reviews with empty or placeholder values (e.g., "", "NA", "N/A", "Unknown") are flagged because they do not contain meaningful textual information.

Duplicate Records

-Duplicate reviews are identified using the "review_id". Only the first occurrence is retained, while subsequent duplicates are flagged to avoid repeated records in downstream analysis.

Non-English Review Content

-Since the project focuses on English-language text analysis, reviews whose detected language is not English are flagged for removal.

Empty Review After Cleaning

-After text preprocessing (such as removing URLs, emojis, punctuation, and other non-textual characters), some reviews may become empty. These records are also flagged because they no longer contain analyzable content.

The quality flags were designed to document why a raw review should be excluded from the **Cleaned_Reviews dataset** while preserving the original record for traceability.

Instead of deleting records immediately, each raw review is evaluated against a series of predefined quality checks. If a record fails a quality check, a removal reason is recorded in the 'record_remove_reason' field. This allows every removed record to remain traceable in the **Raw_Reviews table**.

- **Table relationship**

1. All foreign key relationships remained valid after the ingestion process. No orphan records were detected in any of the three relationships.
2. The database query results are consistent with the Python ETL Statistics:

1931 raw review records did not have corresponding records in Cleaned_Reviews because they were intentionally excluded during the cleaning stage due to non-English content.

The skipped duplicates seem a little bit weird here, since it skips more records than expected; however, the insertion is constantly 0, showing that the skipping logic works successfully for the ingestion runs using the same data.

For data extraction on 7/19/2026 (9:59)

- **Ingestion run one**

Table	Records Inserted	Duplicates	Errors

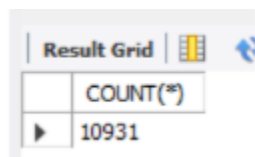
App	0	5	0
Country	0	5	0
Raw_Reviews	147	10784	0
Cleaned_Reviews	115	8779	0

- **Quality Flags**

Sum(condition=True)

Duplicate Records	Missing Review Content	Non-English Review Content	Empty Review After Cleaning
0	0	1963	0

**SELECT COUNT(*)
FROM Raw_Reviews;**



Result Grid	COUNT(*)
▶	10931

Current Raw_Reviews count - Raw_Reviews count in ingestion run 1:

10931-10784=147

This is aligned with the new raw_review record count inserted while using The New RSS data.

**SELECT COUNT(*)
FROM Cleaned_Reviews;**

Result Grid	
	COUNT(*)
▶	8968

Current Cleaned_Reviews count - Cleaned_Reviews count in ingestion run 1:

$$8968-8853=115$$

This is aligned with the new cleaned_review record count inserted while using the New RSS data.

```
SELECT COUNT(*) AS valid_raw
FROM Raw_Reviews
WHERE record_remove_reason IS NULL;
```

Result Grid	
	valid_raw
▶	8968

Valid_raw count =Cleaned_reviews count in the current run

$$8968=8968$$

This suggests that the records count passing quality_flag_check from raw_reviews is aligned with the current cleaned_review record number.

```
SELECT COUNT(*) AS missing_clean
FROM Raw_Reviews r
LEFT JOIN Cleaned_Reviews c
ON r.raw_review_pk = c.raw_review_pk
WHERE r.record_remove_reason IS NULL
AND c.raw_review_pk IS NULL;
```

Result Grid	Filter Rows
missing_clean	
0	

This further suggests that all the cleaned records are put into the cleaned_review table. Combining the information from the former query (the equal count), we can confidently conclude that the filtering between the current raw_reviews table and the current cleaned_reviews table is processed correctly.

```
SELECT raw_review_pk, COUNT(*)
FROM Cleaned_Reviews
GROUP BY raw_review_pk
HAVING COUNT(*) > 1;
```

Result Grid	Filter Rows
raw_review_pk	COUNT(*)

There are no duplicated cleaned raw_reviews records being inserted into the Cleaned_reviews table, which is aligned with the table relationships.

The current ingestion run yielded Non-English Review Content count(Total record_remove_reason count) - That count for the first ingestion run =1963-1931=32

Which is aligned with the current run insertion difference between the two tables
147-115=32

Suggesting the insertion difference between the raw_reviews and cleaned_reviews tables is reasonable, since there is a correct number of newly inserted records without passing quality validations, being prevented from accessing the cleaned_reviews tables.

Based on the query outcomes above, we could confidently enough to claim that The insertion of new data and the skipping of duplicated records while implementing the dynamic data ingestion is good enough for us to carry out the later analysis.

